Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
1  SELECT f.*  344 ms
2  FROM flight f
3  JOIN airline a  1..n<->1: ON f.airline_id = a.airline_id
4  WHERE a.airline_name = 'Air Astana';
5
6  SELECT f.flight_id, a.airport_name
7  FROM flight f
8  JOIN airport a  1..n<->1: ON f.departure_airport_id = a.airport_id;
9
10 SELECT a.*
11 FROM airline a
12 LEFT JOIN flight f ON a.airline_id = f.airline_id
13 AND f.departure_time >= DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '1 month'
14 AND f.departure_time < DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '2 months'
15 WHERE f.flight_id IS NULL;
16
17 SELECT n.*
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 344 ms
    - console 157 ms
    - console_1 369 ms

Output | airport_lab6.public.flight

Tx: Auto | DDL | CSV

	flight_id	airline_id	departure_a…	arrival_air…	departure_time
1	1	1	1	2	2026-11-10 08:00:00.0000
2	2	1	1	3	2026-11-12 10:00:00.0000
3	6	1	2	5	2026-12-15 11:00:00.0000

3 rows

Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
1  SELECT f.*
2  FROM flight f
3  JOIN airline a 1..n<->1: ON f.airline_id = a.airline_id
4  WHERE a.airline_name = 'Air Astana';
5
6  SELECT f.flight_id, a.airport_name  358 ms
7  FROM flight f
8  JOIN airport a 1..n<->1: ON f.departure_airport_id = a.airport_id;
9
10 SELECT a.*
11 FROM airline a
12 LEFT JOIN flight f ON a.airline_id = f.airline_id
13 AND f.departure_time >= DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '1 month'
14 AND f.departure_time < DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '2 months'
15 WHERE f.flight_id IS NULL;
16
17 SELECT n.*
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 358 ms
    - console 157 ms
    - console_1 369 ms

Output | Result 18

Tx: Auto | DDL | CSV

	flight_id	airport_name
1	1	Almaty International Airport
2	2	Almaty International Airport
3	3	Astana International Airpor
4	4	Istanbul Airport

6 rows

```
SELECT f.*
FROM flight f
JOIN airline a 1..n<->1: ON f.airline_id = a.airline_id
WHERE a.airline_name = 'Air Astana';

SELECT f.flight_id, a.airport_name
FROM flight f
JOIN airport a 1..n<->1: ON f.departure_airport_id = a.airport_id;

SELECT a.*  366 ms
FROM airline a
LEFT JOIN flight f ON a.airline_id = f.airline_id
AND f.departure_time >= DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '1 month'
AND f.departure_time < DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '2 months'
WHERE f.flight_id IS NULL;
```

	airline_id	airline_name	home_country
1	2	SCAT Airlines	Kazakhstan
2	5	Qatar Airways	Qatar

Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
12  LEFT JOIN flight f ON a.airline_id = f.airline_id
13  AND f.departure_time >= DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '1 month'
14  AND f.departure_time < DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '2 months'
15  WHERE f.flight_id IS NULL;
16
17  SELECT p.*  369 ms
18  FROM passenger p
19  JOIN booking b  1<->1..n: ON p.passenger_id = b.passenger_id
20  WHERE b.flight_id = 1;
21
22  SELECT f.flight_id,
23         AVG(b.ticket_price),
24         SUM(b.ticket_price),
25         MAX(b.ticket_price),
26         MIN(b.ticket_price)
27  FROM flight f
28  LEFT JOIN booking b  1<->0..n: ON f.flight_id = b.flight_id
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 369 ms
    - console 157 ms
    - console_1 369 ms

Output | airport_lab6.public.passenger

Tx: Auto | DDL | CSV

	passenger_id	first_name	last_name	birth_date	passport_r
1	1	Amina	Sultanova	2004-05-12	KZ123456
2	2	Dias	Nurgali	2008-09-20	KZ234567

2 rows

Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
21
22  SELECT f.flight_id,   395 ms
23         AVG(b.ticket_price),
24         SUM(b.ticket_price),
25         MAX(b.ticket_price),
26         MIN(b.ticket_price)
27  FROM flight f
28  LEFT JOIN booking b  1<->0..n:  ON f.flight_id = b.flight_id
29  GROUP BY f.flight_id;
30
31  SELECT f.flight_id, a.airline_name, ap.country
32  FROM flight f
33  JOIN airline a  1..n<->1:  ON f.airline_id = a.airline_id
34  JOIN airport ap  1..n<->1:  ON f.arrival_airport_id = ap.airport_id
35  WHERE ap.country = 'Kazakhstan';
36
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 395 ms
    - console 157 ms
    - console_1 369 ms

Output | Result 21

csv

	flight…	avg	sum	max	min
1	2	122500	245000	125000	120000
2	3	35000	35000	35000	35000
3	5	180000	180000	180000	180000
4	4	150000	150000	150000	150000

6 rows

Safari

Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
25        MAX(b.ticket_price),
26        MIN(b.ticket_price)
27    FROM flight f
28    LEFT JOIN booking b  1<->0..n:  ON f.flight_id = b.flight_id
29    GROUP BY f.flight_id;
30
31    SELECT f.flight_id, a.airline_name, ap.country   387 ms
32    FROM flight f
33    JOIN airline a  1..n<->1:  ON f.airline_id = a.airline_id
34    JOIN airport ap  1..n<->1:  ON f.arrival_airport_id = ap.airport_id
35    WHERE ap.country = 'Kazakhstan';
36
37    SELECT p.first_name, p.last_name, ap.city, ap.country
38    FROM passenger p
39    JOIN booking b  1<->1..n:  ON p.passenger_id = b.passenger_id
40    JOIN flight f  1..n<->1:  ON b.flight_id = f.flight_id
41    JOIN airport ap  1..n<->1:  ON f.arrival_airport_id = ap.airport_id
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

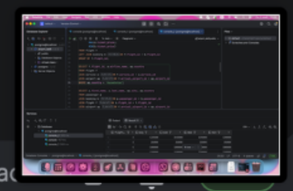
- Database
  - postgres@localhost
    - console_2 387 ms
    - console 157 ms
    - console_1 369 ms

Output | Result 22

Tx: Auto | DDL | CSV

	flight…	airline_na…	country
1	1	Air Astana	Kazakhstan
2	3	SCAT Airlines	Kazakhstan
3	4	Turkish Airlines	Kazakhstan
4	5	Lufthansa	Kazakhstan

4 rows



Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
32  FROM flight f
33  JOIN airline a  1..n<->1: ON f.airline_id = a.airline_id
34  JOIN airport ap  1..n<->1: ON f.arrival_airport_id = ap.airport_id
35  WHERE ap.country = 'Kazakhstan';
36
37  SELECT p.first_name, p.last_name, ap.city, ap.country  358 ms
38  FROM passenger p
39  JOIN booking b  1<->1..n: ON p.passenger_id = b.passenger_id
40  JOIN flight f  1..n<->1: ON b.flight_id = f.flight_id
41  JOIN airport ap  1..n<->1: ON f.arrival_airport_id = ap.airport_id
42  WHERE p.birth_date > CURRENT_DATE - INTERVAL '18 years';
43
44  SELECT CONCAT(p.first_name, ' ', p.last_name) AS full_name,
45         p.passport_number, f.arrival_time
46  FROM passenger p
47  JOIN booking b  1<->1..n: ON p.passenger_id = b.passenger_id
48  JOIN flight f  1..n<->1: ON b.flight_id = f.flight_id;
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 358 ms
    - console 157 ms
    - console_1 369 ms

Output | Result 23

Tx: Auto | DDL | CSV

	first_name	last_name	city	country
1	Aliya	Bekova	Istanbul	Turkey

1 row

Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
42  WHERE p.birth_date > CURRENT_DATE - INTERVAL '18 years';
43
44  SELECT CONCAT(p.first_name, ' ', p.last_name) AS full_name,   378 ms
45         p.passport_number, f.arrival_time
46  FROM passenger p
47  JOIN booking b  1<->1..n: ON p.passenger_id = b.passenger_id
48  JOIN flight f  1..n<->1: ON b.flight_id = f.flight_id;
49
50  SELECT ap.country, a.airline_name, COUNT(*) AS total_flights
51  FROM flight f
52  JOIN airline a  1..n<->1: ON f.airline_id = a.airline_id
53  JOIN airport ap  1..n<->1: ON f.departure_airport_id = ap.airport_id
54  WHERE a.home_country = ap.country
55  GROUP BY ap.country, a.airline_name;
56
57  SELECT f.flight_id, a.airline_name,
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 378 ms
    - console 157 ms
    - console_1 369 ms

Output | Result 24

csv

	full_name	passport_number	arrival_time
1	Amina Sultanova	KZ123456	2026-11-10 09:45:00.000000
2	Dias Nurgali	KZ234567	2026-11-10 09:45:00.000000
3	Aliya Bekova	KZ345678	12 16:00:00.000000
4	Arman Tulegenov	KZ456789	2026-11-15 13:45:00.000000

7 rows

Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
49
50  SELECT ap.country, a.airline_name, COUNT(*) AS total_flights  354 ms
51  FROM flight f
52  JOIN airline a  1..n<->1: ON f.airline_id = a.airline_id
53  JOIN airport ap  1..n<->1: ON f.departure_airport_id = ap.airport_id
54  WHERE a.home_country = ap.country
55  GROUP BY ap.country, a.airline_name;
56
57  SELECT f.flight_id, a.airline_name,
58         d.airport_name AS departure_airport,
59         r.airport_name AS arrival_airport
60  FROM flight f
61  JOIN airline a  1..n<->1: ON f.airline_id = a.airline_id
62  JOIN airport d  1..n<->1: ON f.departure_airport_id = d.airport_id
63  JOIN airport r  1..n<->1: ON f.arrival_airport_id = r.airport_id;
64
65  SELECT p.first_name, p.last_name, f.flight_id
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 354 ms
    - console 157 ms
    - console_1 369 ms

Output | Result 25

CSV

	country	airline_name	total_flights
1	Germany	Lufthansa	1
2	Kazakhstan	Air Astana	3
3	Kazakhstan	SCAT Airlines	1
4	Turkey	Turkish Airlines	1

4 rows

Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
51  FROM flight f
52  JOIN airline a  1..n<->1: ON f.airline_id = a.airline_id
53  JOIN airport ap  1..n<->1: ON f.departure_airport_id = ap.airport_id
54  WHERE a.home_country = ap.country
55  GROUP BY ap.country, a.airline_name;
56
57  SELECT f.flight_id, a.airline_name,  360 ms
58         d.airport_name AS departure_airport,
59         r.airport_name AS arrival_airport
60  FROM flight f
61  JOIN airline a  1..n<->1: ON f.airline_id = a.airline_id
62  JOIN airport d  1..n<->1: ON f.departure_airport_id = d.airport_id
63  JOIN airport r  1..n<->1: ON f.arrival_airport_id = r.airport_id;
64
65  SELECT p.first_name, p.last_name, f.flight_id
66  FROM passenger p
67  JOIN booking b  1<->1..n: ON p.passenger_id = b.passenger_id
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 360 ms
    - console 157 ms
    - console_1 369 ms

Output | Result 26

CSV

	flight_id	airline_name	departure_airport	arrival_airport
1	1	Air Astana	Almaty International Airport	Astana International Airport
2	2	Air Astana	Almaty International Airport	Istanbul Airport
3	3	SCAT Airlines	nternational Airport	Almaty International Airport
4	4	Turkish Airlines	Istanbul Airport	Almaty International Airport

6 rows

Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
59        r.airport_name AS arrival_airport
60  FROM flight f
61  JOIN airline a  1..n<->1: ON f.airline_id = a.airline_id
62  JOIN airport d  1..n<->1: ON f.departure_airport_id = d.airport_id
63  JOIN airport r  1..n<->1: ON f.arrival_airport_id = r.airport_id;
64
65  SELECT p.first_name, p.last_name, f.flight_id   373 ms
66  FROM passenger p
67  JOIN booking b  1<->1..n: ON p.passenger_id = b.passenger_id
68  JOIN flight f  1..n<->1: ON b.flight_id = f.flight_id
69  JOIN airline a  1..n<->1: ON f.airline_id = a.airline_id
70  WHERE a.airline_name = 'Air Astana';
71
72  SELECT f.flight_id, ap.country,
73         COUNT(DISTINCT b.passenger_id) AS passenger_count
74  FROM flight f
75  JOIN airport ap  1..n<->1: ON f.arrival_airport_id = ap.airport_id
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 373 ms
    - console 157 ms
    - console_1 369 ms

Output | Result 27

Tx: Auto | DDL | CSV

	first_name	last_name	flight_id
1	Amina	Sultanova	1
2	Dias	Nurgali	1
3	Aliya	Bekova	2
4	Dias	Nurgali	2

4 rows

Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
WHERE a.airline_name = 'Air Astana';

SELECT f.flight_id, ap.country,   354 ms
       COUNT(DISTINCT b.passenger_id) AS passenger_count
FROM flight f
JOIN airport ap  1..n<->1: ON f.arrival_airport_id = ap.airport_id
LEFT JOIN booking b  1<->0..n: ON f.flight_id = b.flight_id
GROUP BY f.flight_id, ap.country;

SELECT p.*
FROM passenger p
LEFT JOIN booking b  1<->0..n: ON p.passenger_id = b.passenger_id
WHERE b.booking_id IS NULL;

SELECT a.airline_name, COUNT(f.flight_id) AS total_flights
FROM airline a
LEFT JOIN flight f  1<->0..n: ON a.airline_id = f.airline_id
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 354 ms
    - console 157 ms
    - console_1 369 ms

Output | Result 28

csv

	flight_id	country	passenger_count
1	1	Kazakhstan	2
2	2	Turkey	2
3	3	Kazakhstan	1
4	4	Kazakhstan	1

6 rows

Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
76  LEFT JOIN booking b  1<->0..n:  ON f.flight_id = b.flight_id
77  GROUP BY f.flight_id, ap.country;
78
79  SELECT p.*  346 ms
80  FROM passenger p
81  LEFT JOIN booking b  1<->0..n:  ON p.passenger_id = b.passenger_id
82  WHERE b.booking_id IS NULL;
83
84  SELECT a.airline_name, COUNT(f.flight_id) AS total_flights
85  FROM airline a
86  LEFT JOIN flight f  1<->0..n:  ON a.airline_id = f.airline_id
87  GROUP BY a.airline_id, a.airline_name;
88
89  SELECT a.airline_name, f.flight_id,
90         ap.airport_name, ap.city
91  FROM flight f
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 346 ms
    - console 157 ms
    - console_1 369 ms

Output | airport_lab6.public.passenger

Tx: Auto | DDL | CSV

	passenger_id	first_name	last_name	birth_date	passport_r
1	6	Miras	Omarov	2015-06-18	KZ678901

1 row

Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
81  LEFT JOIN booking b  1<->0..n:  ON p.passenger_id = b.passenger_id
82  WHERE b.booking_id IS NULL;
83
84  SELECT a.airline_name, COUNT(f.flight_id) AS total_flights   361 ms
85  FROM airline a
86  LEFT JOIN flight f  1<->0..n:  ON a.airline_id = f.airline_id
87  GROUP BY a.airline_id, a.airline_name;
88
89  SELECT a.airline_name, f.flight_id,
90         ap.airport_name, ap.city
91  FROM flight f
92  JOIN airline a  1..n<->1:  ON f.airline_id = a.airline_id
93  JOIN airport ap  1..n<->1:  ON f.departure_airport_id = ap.airport_id
94  WHERE ap.country = 'Kazakhstan';
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 361 ms
    - console 157 ms
    - console_1 369 ms

Output | Result 30

CSV

	airline_name	total_flights
1	Turkish Airlines	1
2	Qatar Airways	0
3	Lufthansa	
4	SCAT Airlines	1

5 rows

Database Explorer

- postgres@localhost 2 of 3
  - airport_lab6 1 of 3
    - public
    - Database Objects
    - virtual views 1
  - postgres 1 of 3
  - Server Objects

console [postgres@localhost] | console_1 [postgres@localhost] | console_2 [postgres@localhost]

Tx: Auto | Playground | airport_lab6.public

```
81  LEFT JOIN booking b  1<->0..n:  ON p.passenger_id = b.passenger_id
82  WHERE b.booking_id IS NULL;
83
84  SELECT a.airline_name, COUNT(f.flight_id) AS total_flights
85  FROM airline a
86  LEFT JOIN flight f  1<->0..n:  ON a.airline_id = f.airline_id
87  GROUP BY a.airline_id, a.airline_name;
88
89  SELECT a.airline_name, f.flight_id,   356 ms
90         ap.airport_name, ap.city
91  FROM flight f
92  JOIN airline a  1..n<->1:  ON f.airline_id = a.airline_id
93  JOIN airport ap  1..n<->1:  ON f.departure_airport_id = ap.airport_id
94  WHERE ap.country = 'Kazakhstan';
```

Files

- default ~/DataGripProjects/default
- Scratches and Consoles

Services

- Database
  - postgres@localhost
    - console_2 356 ms
    - console 157 ms
    - console_1 369 ms

Output | Result 31

Tx: Auto | DDL | CSV

	airline_name	flight…	airport_name	city
1	Air Astana	1	Almaty International Airport	Almaty
2	Air Astana	2	Almaty International Airport	Almaty
3	SCAT Airlines	3	Ast…onal Airport	Astana
4	Air Astana	6	Astana International Airport	Astana

4 rows